

FINAL PROJECT: Machine Learning

Total Points: 100

Deadline: 2nd May 2026

Instructions:

- All work must be completed on an individual basis.
- **Deliverables:** Submit your well-documented Jupyter Notebook (.ipynb or .py file) and a single exported PDF report. **Unlike regular assignments, the Final Project will be scrutinized more thoroughly.** You must convert the entire notebook into a single PDF file using one of the three methods discussed in the very first demo session. Penalties will be strictly applied if you fail to upload a properly formatted PDF containing the entire notebook. Explanations must be written in Markdown cells, code in Code cells, and all outputs/plots must be clearly visible. Do not submit .zip files.
- **Important:** This final project heavily emphasizes your *written analysis* and *conceptual understanding*. Code without detailed, original explanations will receive minimal credit. The use of generative AI to write your report is strictly prohibited and will be checked.
- **CHOOSE ONE OPTION:** You must select *either* Option 1 *or* Option 2. **Please pick only one. No bonus points will be awarded for attempting both.**

OPTION 1: Deep Dive into Neural Networks (100 Points)

Problem Statement: In this project, you will build a Multi-Layer Perceptron (MLP) from scratch (using a framework like TensorFlow/Keras, PyTorch, or Scikit-Learn) to classify images from the CIFAR-10 dataset.

Note: Because Convolutional Neural Networks (CNNs) were not covered in this course, you are strictly prohibited from using Conv2D or related convolutional layers. You must solve this using traditional dense/linear Neural Networks.

Data Source: Load the CIFAR-10 dataset directly using your chosen framework (e.g., `tf.keras.datasets.cifar10.load_data()`). The dataset contains 32x32 color images across 10 classes.

Part A: Data Preparation and Baseline Model (25 Points)

1. **Preprocessing:** Normalize the pixel values of the images. Since you are using an MLP, you must flatten the 32x32x3 images into 1D arrays (vectors of size 3072) before feeding them into the network.
2. **Baseline Architecture:** Build a simple Neural Network with exactly **one hidden layer** (e.g., 128 neurons) using the Sigmoid activation function and a standard Stochastic Gradient Descent (SGD) optimizer.

3. **Baseline Evaluation:** Train for a reasonable number of epochs. Report the training and testing accuracy, and plot the loss curve.

Part B: Hyperparameter Experimentation (35 Points)

Your goal is to achieve the best possible accuracy on the test set by experimenting with the network's architecture and hyperparameters. You must systematically test and document the following three experiments:

1. **Activation Functions & Depth:** Build a "deep" network (at least 4 hidden layers). Compare its performance when using all Sigmoid activations versus all ReLU activations.
2. **Optimizers:** Keep the all ReLU architecture (Ignore Sigmoid) and compare the standard SGD optimizer against the Adam optimizer.
3. **Learning Rate (LR):** Using your best optimizer, test three vastly different learning rates (e.g., a very high LR like 0.1, a standard LR like 0.001, and a very low LR like 0.00001).

Part C: Conceptual Analysis & Written Report (40 Points)

This is the most critical part of the assignment. Answer the following questions in detailed Markdown cells based on your experiments in Part B.

1. **Activation Function Analysis:** Analyze your results from experiment B1. Compare the performance of the network when using Sigmoid activations versus ReLU activations. Based on your loss curves and accuracy, why do you think one performed better than the other in this specific architecture? Explain your reasoning.
2. **Optimizer Mechanics:** Explain why the Adam optimizer behaved differently than SGD in your experiment (B2). What specific mechanisms does Adam use that standard SGD lacks?
3. **Learning Rate Dynamics:** Analyze your loss curves from experiment B3. Describe the visual symptoms of a learning rate that is too high versus one that is too low.
4. **Final Conclusion:** What was your highest achieved test accuracy? Explain why image classification is inherently difficult for flattened 1D arrays and standard MLPs compared to spatial algorithms.

OPTION 2: The Comprehensive ML Pipeline (100 Points)

Problem Statement: You have been hired as a Data Scientist for a medical research facility. Your task is to build a complete, end-to-end Machine Learning pipeline to classify breast cancer tumors as either malignant or benign based on computed features from digitized images.

Data Source: You will use the Breast Cancer Wisconsin (Diagnostic) dataset, which is readily available in scikit-learn. Load it using `sklearn.datasets.load_breast_cancer()`. All features are numerical and continuous.

Part A: Data Preprocessing & Exploratory Analysis (20 Points)

1. Load the dataset and separate it into features (X) and the target variable (y).
2. Standardize/Scale the numerical features. (Explain briefly why scaling is critical for the models you will use later in the pipeline).
3. Create at least two visualizations that explore the relationships between the features. For example, you might create a scatter plot of two features (colored by the target variable) or a correlation heatmap. Briefly explain any trends or separability you observe.

Part B: Unsupervised Learning & Dimensionality Reduction (25 Points)

1. **PCA:** Drop the target variable for this step. Apply Principal Component Analysis (PCA) to the standardized feature set. How many components are required to explain at least 90% of the variance in the data?
2. **Clustering:** Apply K-Means clustering (using $K = 2$, since we know there are two true tumor classes) to the standardized dataset. Create a scatter plot of the resulting clusters, using the first two Principal Components as your X and Y axes.
3. **Analysis:** Compare your K-Means cluster labels to the actual target labels (Malignant vs. Benign). Did the clustering algorithm naturally group the malignant and benign tumors together without knowing the true labels? Explain your observations.

Part C: Supervised Learning Comparison (30 Points)

Split your standardized dataset into a 70-30 Train/Test split. Train the following three classifiers to predict whether a tumor is malignant or benign:

1. **Logistic Regression**
2. **Support Vector Machine (SVM):** Use an RBF kernel.
3. **Decision Tree Classifier:** Limit the max_depth parameter to prevent severe overfitting.

Part D: Model Evaluation and Real-World Interpretation (25 Points)

Answer the following in detailed Markdown cells.

1. **Metrics:** For all three models, calculate and report the Accuracy, Precision, Recall, and F-1 Score on the test set.
2. **ROC Curve:** Plot the ROC curves for all three models on a single graph and report their AUC (Area Under the Curve) scores.
3. **Medical Decision:** In the context of predicting Malignant vs. Benign tumors, which metric is more critical: Precision or Recall? (Hint: Think deeply about the real-world cost of a False Negative—missing a malignant tumor—versus the cost of a False Positive—a false alarm).
4. **Final Recommendation:** Based on your metrics, the ROC curve, and your answer to Question 3, which of the three models would you deploy to assist doctors, and why? Be specific about the trade-offs you considered.